



BYRON KHARL B. AGA-IN

Sta.Catalina, Lubao, Pampanga

Phone: 09654189950

Email: [Email Address] • LinkedIn/GitHub: [Links]

Professional Summary

Third-year Bachelor of Science in Information Technology (BSIT) student with hands-on experience in Internet of Things (IoT) development, embedded systems, and artificial intelligence. Adept at bridging hardware prototyping with low-level software optimization and computer vision applications. Strong foundation in electronic hardware design, complex debugging, and project lifecycle management.

Education

Bachelor of Science in Information Technology (BSIT) – 3rd Year

[University Name] | Angeles, Philippines | Expected Graduation: 2027

Technical Skills

Hardware & Embedded Systems:	ESP32, ATmega 328P, ESP-01, IoT Sensors (pH, DO, Temp, Turbidity)
Programming Languages:	8086 Assembly (TASM), C/C++
Software & Tools:	KiCad (PCB Design), Convolutional Neural Networks (CNNs)
Project Management:	Technical documentation, Gantt charting, timeline planning

Academic Projects

AI-Based Aquaculture Disease Detection System

Lead Developer / Researcher

- Developing an early warning system leveraging Convolutional Neural Networks (CNNs) to detect and classify catfish diseases via image processing.
- Structuring comprehensive research documentation, including problem statements and objectives, targeting backyard aquaculture systems in the Pampanga region.

Automated IoT Water Quality Monitor

Hardware & Systems Designer

- Designing an automated monitoring device for fish ponds to track real-time pH, dissolved oxygen, temperature, and turbidity levels.
- Managing end-to-end electronic hardware prototyping, encompassing component integration, 3D prototype modeling, and system evaluation.

- Coordinating a 12-month engineering timeline from initial planning to final reporting using detailed Gantt charts.

Low-Level Algorithm Optimization

Programmer

- Built and optimized sorting algorithms utilizing low-level 8086 Assembly language and TASM.
- Debugged complex logic issues, register errors, and operand mismatches to ensure highly efficient code execution.

PCB Prototyping and Design

Hardware Engineer

- Designed custom Printed Circuit Boards (PCBs) utilizing KiCad software.
- Troubleshoot and resolved Design Rule Check (DRC) errors, including board outline corrections, copper zone/ground plane implementation, and precise footprint modifications.